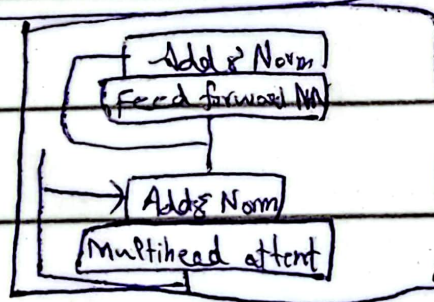


Transformer:- (Encoder):-

In a transformer there are 2 blocks encoder & decoder.

- > In encoder there is 8 identical blocks
- In each encoder there is 2 blocks

Feed forward N.N
self attention



→ 1 encode block
Architecture.

Input

Date: _____

Day: _____

How:- Sentence How are you.

→ In input we do 3 operations.

- 1) Tokenization (Divide sentence to parts).
- 2) Embedding (Convert to vectors like OHE) means nos.
- 3) Positional encoding. (Add pos of word).

→ Then we applied multihead attention.

→ Above it do residual connection we do Layer normalization.

→ We send it to feed forward & F.F. done their processes.

→ Again residual add & Layer norm.

→ Output we send to other encoder.

Repeat this 6 times.

Why use residual connection:-

→ "ف" "ن" "و" "س"

seq to seq $\rightarrow$ Autoregressive as dep on prev output.
non-auto $\rightarrow$ Not possible. (as you haven't prev data)

Date: _____

Day:_____

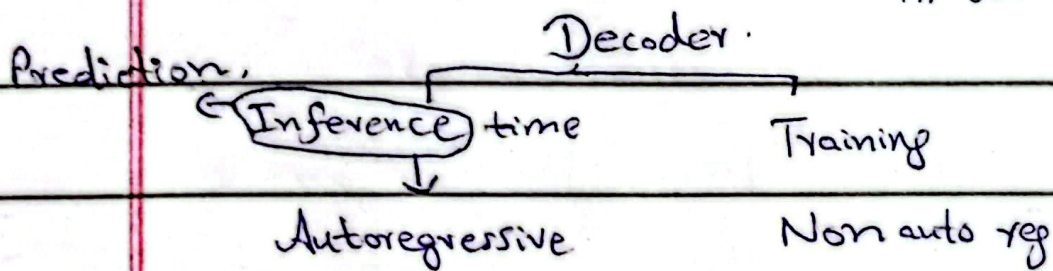
Masked self attention:-

Autoregressive models:-

Transformer decoder is auto-regressive ^{time-series}

at inference time & non-auto-regressive
at training time.

Table of teacher forces
im don't need. (as exact sentence
will be typed)



autoregressive:- At any time step if u have to predict output, u need output from previous time step

→ Main problem here is the data leakage when u use non-auto behaviour at training time as it ^{data} will go in parallel & in simple its cheating & never find results at prediction time

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Non - slow $\Rightarrow$ $O(n^2)$ $\Rightarrow$ aggressive $\Rightarrow$

- cheating - $\Rightarrow$ $O(n^2)$ $\Rightarrow$

Answer is we do masking.

$\rightarrow$ Add a matrix in self attention with 0's $\&$ $-\infty$ with the 1 u don't want $\&$ then pass to softmax.

$\rightarrow$ $\&$ softmax $(-\infty) = 0$.

$\rightarrow$ AND through this masking solve the problem. (Data doesn't leak) $\&$ achieved non-auto reg model at training time in which data will go in parallel that's why will be very fast.